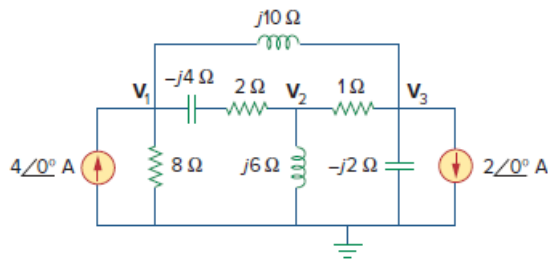


## Problem

Determine  $V_1$ ,  $V_2$ , and  $V_3$  in the circuit of Fig. 10.129 using *PSpice* or *MultiSim*.

Figure 10.129:



## Step-by-step solution

## Step 1 of 3

The schematic is shown below. We insert three pseudo component prints at

Nodes 1, 2 and 3, we set Total pts = 1, Start frequency = 0.1592 and the End frequency = 0.1592. In the AC sweep box. After simulation, the output file includes,

|                 |                  |                  |
|-----------------|------------------|------------------|
| Frequency       | $V M(\$M\_0004)$ | $V P(\&N\_0004)$ |
| $1.592 E^{-01}$ | $1.591 E^{+01}$  | $1.696 E^{+02}$  |
| Frequency       | $V M(\$M\_0001)$ | $V P(\&N\_0001)$ |
| $1.592 E^{-01}$ | $5.172 E^{+00}$  | $-1.386 E^{+02}$ |
| Frequency       | $V M(\$M\_0003)$ | $V P(\&N\_0003)$ |
| $1.592 E^{-01}$ | $2.270 E^{+00}$  | $-1.524 E^{+02}$ |

## Step 2 of 3

Therefore,

$$\begin{aligned} V_1 &= 15.91 \angle 169.6^\circ V, \\ V_2 &= 5.172 \angle -138.6^\circ V \quad \& \\ V_3 &= 2.27 \angle -152.4125.4^\circ V. \end{aligned}$$

## Step 3 of 3

